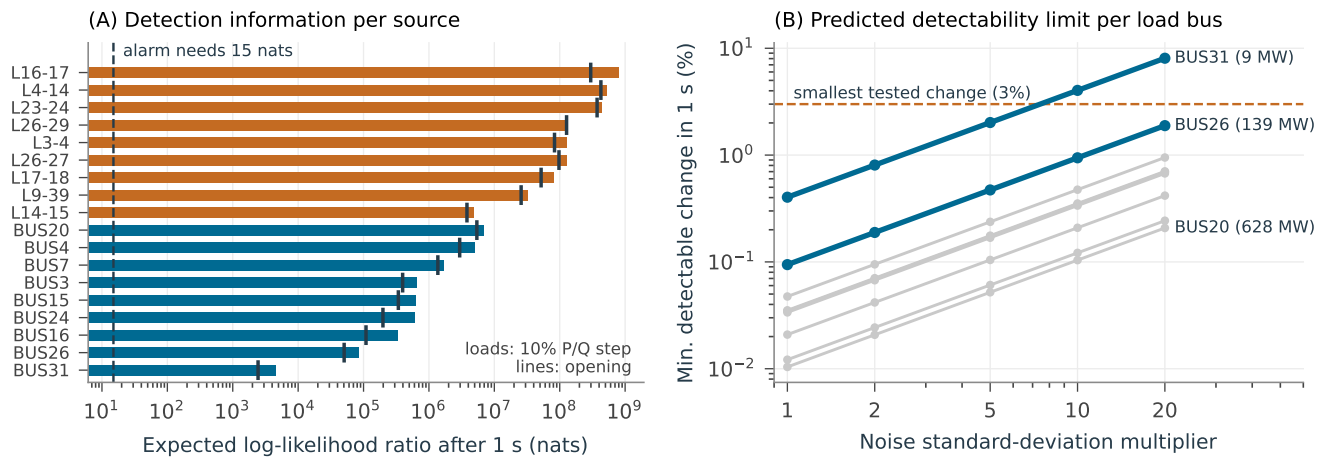
(A) Detection information per source
alarm needs 15 nats
L16-17
L4-14
L23-24
L26-29
L3-4
L26-27
L17-18
L9-39
L14-15
BUS20
BUS4
BUS7
BUS3
BUS15
BUS24
BUS16
BUS26
BUS31
loads: 10% P/Q step
lines: opening
10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8 10^9
Expected log-likelihood ratio after 1 s (nats)
(B) Predicted detectability limit per load bus
smallest tested change (3%)
BUS31 (9 MW)
BUS26 (139 MW)
BUS20 (628 MW)
Min. detectable change in 1 s (%)
10^1
10^0
10^{-1}
10^{-2}
1 2 5 10 20
Noise standard-deviation multiplier